

Executive Summary: Nassau Candy Distributor - Factory Reallocation and Shipping Optimization

Overview

This project introduces a decision-support system that evaluates factory reassignment scenarios before operations teams make allocation changes. The objective is to reduce delivery lead time while maintaining profitability and avoiding unnecessary factory reassignments.

Key Findings

- The system can evaluate multiple factory-product-region scenarios and estimate their likely operational impact before implementation.
- Several existing factory assignments were already optimal, which demonstrates the value of a recommendation engine that can explicitly advise against unnecessary reassignment.
- Selected scenarios achieved measurable lead-time improvement, with observed gains ranging from approximately 2% to 20%.
- The recommendation logic balances speed and profitability, producing more practical decisions than a single-metric optimization approach.

Business Impact

Project 3 reduces the operational risk of trial-and-error reassignment. Instead of changing factory allocations reactively, managers can evaluate the likely service and profit effect in advance and prioritize only the scenarios with credible business benefit.

Strategic Recommendations

- Integrate reassignment scenario evaluation into routine logistics planning rather than using static factory rules alone.
- Pilot the strongest candidate recommendations in controlled operating environments before full-scale rollout.
- Track realized lead-time and profitability outcomes to strengthen confidence in future recommendations.
- Refresh the predictive engine with updated shipment data so the decision-support system remains operationally relevant.

Conclusion

Project 3 provides Nassau Candy Distributor with a scalable foundation for proactive logistics planning. By combining prediction with scenario evaluation, the business can improve shipping decisions while reducing avoidable operational disruption.